



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

FULL SYLLABUS

12-02-2023

JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-II)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,C,D	A,B,C	A,B,C	A,B,C	A,C,D	B,C,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	D	C	D				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	0.94 to 0.95	1.50	3.75	4.25	2.33	1.35	6.00	0.09 to 0.10

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,C	B,C	A,B,D	B,D	A,B,D	A,B,C,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	C	D	A	B				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	5.00	19.00	13.00	47.62	14.00	6.00	2.00	8.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	B,C	A,C	B,C	A,B	A,B,D	A,B,C		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	B	D	C	B				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	3.00	4.55 or 4.56	5.00	85.00	2.80	2.50	1.25	2.50

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A,C,D)

(A) Any line passing through origin in a T-V graph represents constant pressure. The slope of this line is $\frac{T}{V}$ which is directly proportional to pressure P.

During process DA the slope of line through origin increases in slope indicating increase in pressure.

Also the line joining origin and point C also intersects DA at a point.

$$(B) Q_{AB} = nC_p \Delta T = \frac{5}{2} R(T_B - T_0) = \frac{5}{2} RT_0$$

$$Q_{BC} = nRT \ln \frac{V_C}{V_B} = R2T_0 \ln 2$$

$$Q_{CD} = nC_v \Delta T = -\frac{3R}{2} T_0$$

$$Q_{DA} = -RT_0 \ln 4 = -2RT_0 \ln 2$$

$$(C) \eta = \frac{\text{Work done by gas}}{\text{Heat absorbed}}$$

$$W_{\text{cycle}} = W_{AB} + W_{BC} + W_{CD} + W_{DA}$$

$$= RT_0 + 2RT_0 \ln 2 + 0 - 2RT_0 \ln 2 = RT_0$$

$$\eta = \frac{RT_0}{\frac{5}{2}RT_0 + 2RT_0 \ln 2} = \frac{2}{5 + 4 \ln 2}$$

(D) Maximum slope of T-V line is for process AB

$$P = nR \frac{T}{V}$$

$$P_{\text{max}} = \frac{RT_0}{V_0}$$

2. Ans (A,B,C)

$$C_{eq} \text{ (before any attempt)} = \frac{2}{5}C$$

$$C_{eq1} = \frac{4}{9}C \text{ and } C_{eq2} = \frac{6}{11}C$$

$$\Delta C_1 = 1st \text{ change} = \frac{4}{9}C - \frac{2}{5}C = \frac{2}{45}C$$

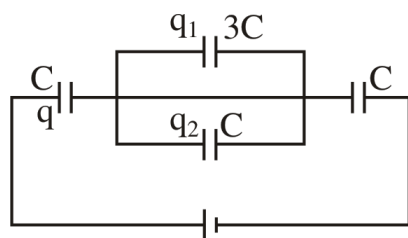
$$\Delta C_2 = 2nd \text{ change} = \frac{6}{11}C - \frac{2C}{5} = \frac{8}{55}C$$

$$(A) \frac{2}{45}C : \frac{8}{55}C = \frac{11}{36}$$

$$(B) W_{cell} = \Delta q \times emf = \Delta C \times \varepsilon^2$$

So ratio of work done is same as ratio of option (A)

(C) After attempt-1 circuit becomes



$$q_1 = \frac{3}{4}q$$

$$\frac{U_A}{U_B} = \frac{q^2}{2C} \bigg/ \frac{\left(\frac{3}{4}q\right)^2}{6C} = 16 : 3$$

(D) Heat dissipated cannot be calculated as external

work is involved

3. Ans (A,B,C)

The ring in horizontal plane is cutting the flux

due to time varying component of $\vec{B}$

$$emf \ \varepsilon = \pi R^2 \frac{d}{dt} (B_0 t^2) = 2\pi R^2 B_0 t$$

$$i = \frac{2\pi R^2 B_0 t}{\lambda 2\pi R} = \frac{RB_0 t}{\lambda}$$

'M' magnetic moment due to this current

$$\pi R^2 \frac{RB_0 t}{\lambda}. \text{ The direction of M is negative y}$$

according to Renz's law.

This is perpendicular to $t_0 B_0 \hat{i}$, so a torque

starts acting having direction $-\hat{j} \times \hat{i} = \hat{k}$.

This torque is balanced by the supports so

reaction at S_1 increases and S_2 decreases (B).

Also net force is only gravitational so (A).

Let reaction at S_1 and S_2 be N_1 & N_2

$$(N_1 - N_2)R = \tau_B = \frac{\pi R^3 B_0^2 t}{\lambda}$$

$$N_1 - N_2 = \frac{\pi R^2 B_0^2 t}{\lambda} \text{ (C)}$$

It looses contact from S_2 when torque of

gravity balances the magnetic torque about S_1

$$2mgR = \frac{\pi R^3 B_0^2 t}{\lambda} \rightarrow t = \frac{2mg\lambda}{B_0^2 \pi R^2}$$

4. Ans (A,B,C)

Electric field in the cavity is given by $\vec{E} = \frac{\rho \vec{b}}{3\epsilon_0}$

where $\vec{b}$ is the vector from center of outer sphere

to center of cavity.

Here $E = \rho \frac{(1-\eta)R}{3\epsilon_0}$ for $\eta \rightarrow 0$, $E = \frac{\rho R}{3\epsilon_0}$

for $\eta = \frac{1}{2}$, $E = \frac{\rho R}{6\epsilon_0}$ directed along BD

So, $2\eta R = \frac{1}{2}at^2$, where $a = \frac{qE}{m}$

$$t = \sqrt{\frac{12\epsilon_0 m}{\rho q}}$$

(C) It strikes at D

Along y $t = \frac{\eta R}{v_1}$

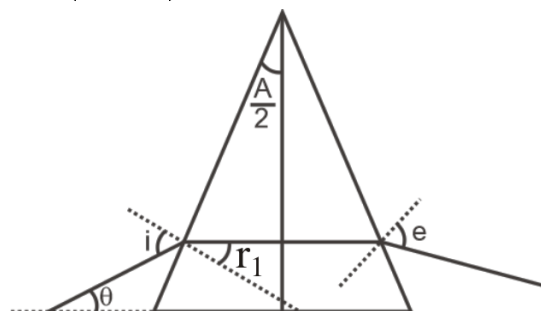
Along x

$$\frac{1}{2}at^2 = \eta R \rightarrow \frac{1}{2}\rho \frac{R}{6\epsilon_0} \frac{q}{m} \frac{\eta^2 R^2}{v_1^2} = \eta R$$

$$v_1 = \frac{R}{2} \sqrt{\frac{\rho q}{6\epsilon_0 m}}$$

$$(D) \Delta K.E = W = qE \frac{R}{2} = q \frac{\rho R}{6\epsilon_0} \frac{R}{2} = \frac{\rho q R^2}{12\epsilon_0}$$

5. Ans (A,C,D)



From geometry $i = \theta + \frac{A}{2}$

For prism-1

$$i = 15^\circ + 30^\circ = 45^\circ$$

By snell's Law at 1st surface

$$\sin 45^\circ = \sqrt{2} \sin r_1,$$

$$r_1 = 30^\circ \rightarrow r_2 = 30^\circ$$

$$\text{So } e = 45^\circ$$

$$\delta = i + e - A = 30^\circ$$

As incident ray is same and emergent rays are parallel this suggests that for each prism condition for minimum deviation is satisfied.

for prism -2

$$i = 15^\circ + 15^\circ = 30^\circ \text{ and } r = \frac{A}{2} = 15^\circ$$

$$\text{So } n_2 = \frac{\sin 30^\circ}{\sin 15^\circ} = 2 \cos 15^\circ = 2 \cos\left(\frac{\pi}{24}\right)$$

for prism-3

$$i = 15^\circ + 45^\circ = 60^\circ \text{ and } r = \frac{A}{2} = 30^\circ$$

$$\text{So } n_3 = \frac{\sin 60^\circ}{\sin 30^\circ} = \sqrt{\frac{3}{2}}$$

6. Ans (B,C,D)

Let mass of rim be equal to m

Applying NLM to wheel

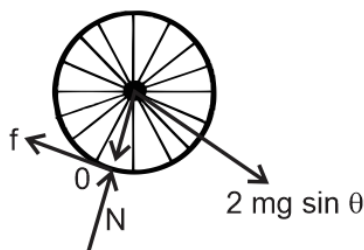
$$2mg \sin \theta - f = 2ma \dots (1)$$

In the C frame of wheel

$$\tau = I\alpha$$

$$fR = \left(mR^2 + \frac{mR^2}{3} \right) \alpha$$

$$f = \frac{4}{3}ma$$



Solving (1) and (2)

$$a = \frac{3}{5}g \sin \theta \rightarrow f = \frac{4}{3} \times \frac{3}{5}mg \sin \theta = \frac{4}{5}mg \sin \theta$$

But

$$f \leq \mu N \rightarrow \frac{4}{5}mg \sin \theta \leq \mu 2mg \cos \theta \rightarrow \mu \geq \frac{2}{5} \tan \theta$$

At a given moment all spokes are in pure rotation

about point 'O' of contact. So ratio of KE will be

the ratio of moment of inertia about 'O'.

$$KE_L : KE_H = \frac{mR^2}{3} : \left(\frac{mR^2}{12} + m \left(\frac{3R}{2} \right)^2 \right)$$

$$= 1 : 7$$

$$\frac{dL}{dt} = \tau = \tau_g = 5g \sin \theta \times R$$

$$= 5 \times 10 \times \frac{1}{2} \times 0.4 = 10 \text{ kg m}^2 \text{ s}^{-1}$$

PART-1 : PHYSICS

SECTION-I (ii)

7. Ans (A)

$$(P) \text{ Magnetic moment} = iA = i(\pi R^2)$$

$$= (7) \left(\frac{22}{7} \right) \left(\frac{5}{100} \times \frac{5}{100} \right)$$

$$= 110 \times 5 \times 10^{-4} = 0.0550$$

$$(Q) \tau = M \times B$$

$$= (0.0550)(100) \times 10^{-3} = 0.0055$$

$$(R) \tau = I\alpha$$

$$0.0055 = \frac{1}{2}(10) \left(\frac{5}{100} \times \frac{5}{100} \right) \alpha$$

$$11 \times 10^{-4} = 25 \times 10^{-4} \alpha$$

$$\alpha = \frac{11}{25} = 0.44$$

$$(S) KE = MB = 0.0055 \text{ J}$$

8. Ans (D)

$$(P) i_{\text{steady}} = \frac{10+5}{6+3} = \frac{15}{9} \text{ mA}$$

$$\Delta V = 10 - 3 \left(\frac{15}{9} \right) = 5 \text{ V olt}$$

$$q_{\text{steady}} = C\Delta V = \frac{1}{5} \times 5 = 1 \mu\text{C}$$

$$(Q) q = 1 \mu\text{C} (1 - e^{-t/RC})$$

$$R = 3M\Omega + \frac{6 \times 3}{6+3} M\Omega = 5M\Omega$$

$$RC = 1$$

$$q = 1 \mu\text{C} (1 - e^{-t})$$

$$q = 1 \mu\text{C} (1 - e^{-1}) = 0.63 \mu\text{C}$$

$$(R) i = \frac{dq}{dt} = +1 \mu\text{A} e^{-t}$$

$$\text{at } t = 1 \text{ s } \quad i = 0.37 \mu\text{A}$$

(S) By Kirchhoff's law

$$5 + 10 - 6(i - 0.37) - 3i = 0$$

$$6(0.37) + 15 = 9i$$

$$i = 1.91 \mu\text{A}$$

$$\text{Current in } 5 \text{ V cell} = 1.54 \mu\text{A}$$

9. **Ans (C)**

$$\vec{V}_I = 2\vec{V}_m - \vec{V}_0$$

$$(P) \vec{V}_I = 2V - 2V = 0$$

$$(Q) V_{I/M} = V_I - V_M = 0 - V$$

$$(R) V_{I/0} = V_I - V_0 = -2V$$

(S) When spring max compressed

$$V_m = 0$$

$$V_I = -V_0 = 2V$$

10. **Ans (D)**

$$(P) 2B \rightarrow C$$

$$Q = BE_C - 2BE_B$$

$$= 60 \times 8.7 - 2(30 \times 8.1) = 36 \text{ MeV}$$

$$(Q) 2B + C \rightarrow D$$

$$Q = 120(8.4) - 2(30 \times 8.1) - 8.7 \times 60 = 0$$

$$(R) E \rightarrow C + D$$

$$Q = (60 \times 8.7) + (120 \times 8.4) - (180 \times 8) = 90 \text{ MeV}$$

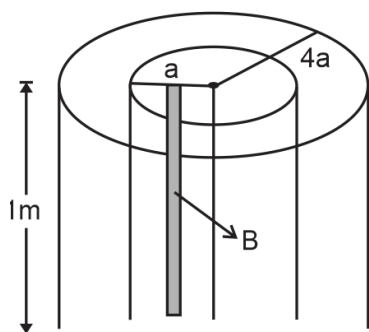
$$(S) 3A \rightarrow C$$

$$Q = (60 \times 8.7) - 3(7 \times 20) = 102 \text{ MeV}$$

PART-1 : PHYSICS

SECTION-II

1. **Ans (0.94 to 0.95)**



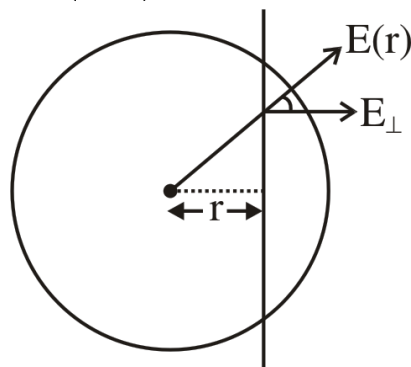
$$\int_0^a \left(\frac{\mu_0 J r}{2} \right)^2 \frac{\mu_0 \pi r dr}{2\mu} + \int_a^{4a} \frac{1}{2} \left(\frac{\mu_0 J a^2}{2r} \right) \frac{1}{\mu_0} 2\pi r dr$$

$$\Rightarrow \frac{\mu_0 J^2 \pi a^4}{4} + \frac{\mu_0 J^2 a^4 \pi}{4} \ln 4 = \frac{1}{2} L J^2 \pi^2 a^4$$

$$\Rightarrow \frac{\mu_0}{\pi} \left[\frac{1}{8} + \frac{\ln 4}{2} \right] = L$$

$$\Rightarrow \frac{\mu_0}{\pi} [0.82] = L$$

2. **Ans (1.50)**



$$E(r) = \frac{\rho r}{3\epsilon_0}, E_{\perp} = \frac{\rho r \cos \theta}{3\epsilon_0} = \frac{\rho x}{3\epsilon_0}$$

$$\phi = \frac{\rho x}{3\epsilon_0} \times \pi (R^2 - x^2)$$

$$= \frac{\rho \pi}{3\epsilon_0} (R^2 x - x^3)$$

$$\text{For maximum } \phi, \frac{d\phi}{dx} = 0$$

$$R^2 - 3x^2 = 0 \rightarrow x = \frac{R}{\sqrt{3}}$$

$$\text{So, } \phi = \frac{\rho \pi}{3\epsilon_0} \left(R^2 \frac{R}{\sqrt{3}} - \frac{R^3}{3\sqrt{3}} \right)$$

$$\text{Putting } R = \sqrt{3}$$

$$\phi_{\max} = 2 \frac{\rho \pi}{3\epsilon_0} = \frac{\rho \pi}{\left(\frac{3}{2}\right) \epsilon_0}$$

$$N = 1.5$$

3. **Ans (3.75)**

For convex lens

$$v_1 = \frac{-36f}{-36 + f}$$

distance of image from plain mirror

$$d = \frac{36f}{-f + 36} - 100 = \frac{36f + 100 - 3600}{36 - f}$$

$$= \frac{136f - 3600}{36 - f}$$

for concave lens

$$u = \frac{136f - 3600}{36 - f} - 65 = \frac{201f - 5940}{36 - f}$$

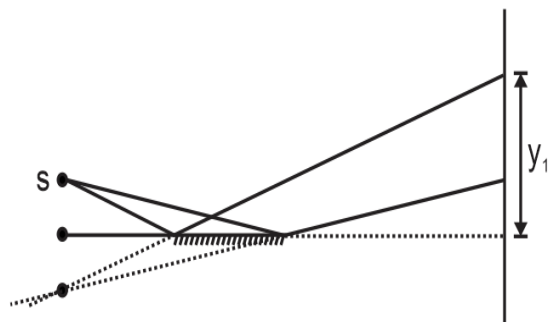
$$\frac{1}{20} - \frac{36 - f}{201f - 5940} = \frac{1}{-2f}$$

$$\Rightarrow f = 30 \text{ cm}$$

Now from lens maker formula $\mu_1 = 2.5$ and μ_2

$$= 1.5$$

4. Ans (4.25)



$$\beta = \frac{\lambda D}{d} = \frac{500 \times 10^{-9} \times 1}{1 \times 10^{-3}} = 500 \times 10^{-6} = 0.5 \text{ mm}$$

$$\frac{y_1}{90} = \frac{0.5 \text{ mm}}{10}$$

$$y_1 = 4.5 \text{ mm}$$

For maxima

$$y \frac{d}{D} = (2N + 1) \frac{\lambda}{2}$$

$$y = (2N + 1) \frac{\lambda D}{2d}$$

$$\frac{\lambda D}{d} = 0.5 \text{ mm}$$

$$y = \frac{(2N + 1)(0.5 \text{ mm})}{2}$$

$$2N + 1 = 17$$

$$y = \left(\frac{17}{2} \right) (0.5 \text{ mm})$$

$$= (8.5)(0.5)$$

$$= 4.25 \text{ mm}$$

5. Ans (2.33)

$$v_0 = \sqrt{\frac{2GM}{R}}$$

By energy conservation

$$\frac{1}{2}mv_0^2 - \frac{GMm}{R} = \frac{1}{2}mv^2 - \frac{GMm}{r}$$

$$v = \sqrt{\frac{2GM}{r}}$$

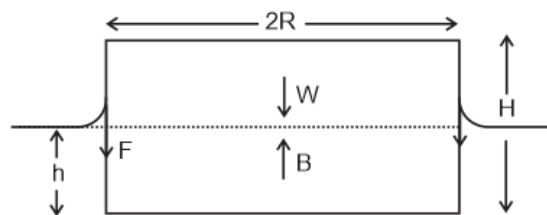
$$\int_{r=R}^{r=4R} \sqrt{r} \, dr = \sqrt{2GM} \int_0^t dt$$

$$t = \frac{7}{3} \sqrt{\frac{2R^3}{GM}}$$

6. Ans (1.35)

$$\frac{E_0 d}{e \ell} = 1.35 \text{ V}$$

7. Ans (6.00)



The disk is in equilibrium under forces of gravity, buoyant force and surface tension.

$$W + F = B$$

$$\pi R^2 H 0.5 \rho_w g + S 2\pi R = \pi R^2 h \rho_w g$$

$$14 \times 14 \times 10 \times 10^{-9} \times 0.5 \times 10^3 \times 10 + 0.07 \times 2 \times 14 \times 10^{-3} = 14 \times 14 \times h \times 10^{-9} \times 10^3 \times 10$$

$$98 \times 10^{-4} + 1.96 \times 10^{-3} = 196 \times 10^{-5} h$$

$$98 \times 10^{-4} + 19.6 \times 10^{-4} = 19.6 \times 10^{-4} h$$

$$h = 6 \text{ mm}$$

8. Ans (0.09 to 0.10)

$$a = \frac{v_2 - v_1}{t} = \frac{0.85 - 0.21}{8.0} = 0.08 \text{ m/s}^2$$

$$\frac{\Delta a}{a} = \frac{\Delta(v_2 - v_1)}{(v_2 - v_1)} + \frac{\Delta t}{t} = \frac{\Delta v_2 + \Delta v_1}{v_2 - v_1} + \frac{\Delta t}{t}$$

$$\frac{\Delta a}{a} = \frac{0.05 + 0.05}{0.85 - 0.21} + \frac{0.1}{8.0}$$

$$= \frac{0.1}{0.64} + \frac{0.1}{8.0}$$

$$\Delta a = 0.08 \left(\frac{0.1}{0.64} + \frac{0.1}{8.0} \right) = 0.0125 + 0.001$$

$$= 0.0135$$

Since error is in the 2nd place of decimal so Δa

must be rounded to 0.0. So acceleration will be

quested as 0.08 ± 0.01

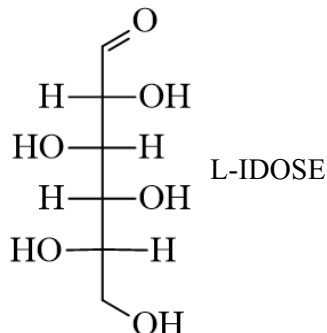
$$|a| + |\Delta a| = 0.09$$

PART-2 : CHEMISTRY

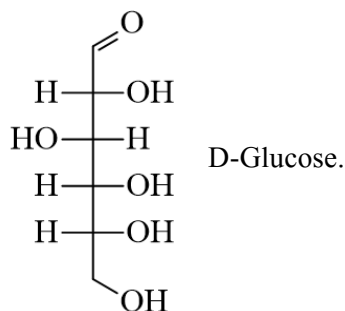
SECTION-I (i)

1. **Ans (A,B,C)**

Unit I acyclic.



Unit II acyclic.



2. **Ans (B,C)**

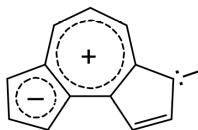
$$\Delta S_{\text{reaction}} = 2 \times 200 - 190 - 3 \times 150 = -240 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$\Delta G = \Delta H - T\Delta S = -99 - \frac{300(-240)}{1000} = -27 \text{ kJ mol}^{-1}$$

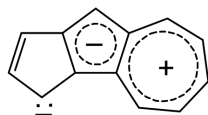
$$\text{Also, } \Delta G = -T\Delta S_{\text{universe}} = \frac{+27 \times 10^3}{300} = +90 \text{ J mol}^{-1} \text{ K}^{-1}$$

3. **Ans (A,B,D)**

(A) The conjugate bases are



More stable due to the alignment of the polarity



(B) EDG and EWG distance increases $\Rightarrow \mu$ increases

(C) Benzene has high resonance energy

(D) SIP lowers basic strength and SIR increases basic strength.

4. **Ans (B,D)**

FACTUAL

5. **Ans (A,B,D)**

As A(l) vaporises, the pressure starts building up and the valve open as soon as the total pressure increases above 1.2.

As, the valve open, He(g) and A(g) diffuses into atmospheric and atmospheric gases diffuse in.

The partial pressure of A(g) will eventually reached to 0.5 atm and the equilibrium would be attained when $P_{\text{total}} = 1.2 \text{ atm}$.

PART-2 : CHEMISTRY

SECTION-I (ii)

7. **Ans (C)**

NCERT THEORY

8. **Ans (D)**

$$\text{Energy of Photon} = \frac{12400}{\lambda(\text{\AA})} \text{ eV}$$

Energy of a particle, E is given by :

$$\lambda_d = \frac{h}{\sqrt{2mKE}}$$

$$\text{For electron } \lambda_d \approx \sqrt{\frac{150}{E(\text{in eV})}} \text{ \AA}$$

(P) 1 Angular node $\Rightarrow$ p orbital

Graph is of 2p.

$$\therefore |\text{energy}| = +13.6 \times \frac{1}{2^2} = 3.4 \text{ eV}$$

9. **Ans (A)**

Working range of the indicator $\text{pH} = \text{pK}_{\text{In}} +$

$$\log \frac{(\text{In}^-)}{(\text{HIn})}$$

For red colour : $\frac{(\text{In}^-)}{(\text{HIn})} \geq 2$ and blue colour :

$$\frac{(\text{In}^-)}{(\text{HIn})} \leq \frac{1}{10}$$

$$\therefore \text{pH} \in [9 - 1, 9 + \log 2] \Rightarrow \text{pH} \in [8, 9.3]$$

(p) Equivalence $\text{pH} = 7$ and end point $\text{pH} \Rightarrow 9.3$ at which colour changes

(q) Equivalence $\text{pH} = 9$ and end point $\text{pH} \Rightarrow 9.3$

(r) Equivalence $\text{pH} = 9$ and end point $\text{pH} \Rightarrow 8$

(r) Equivalence $\text{pH} = 7$ and end point $\text{pH} \Rightarrow 8$

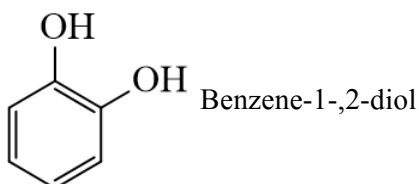
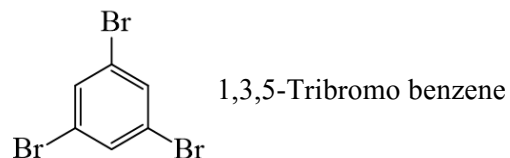
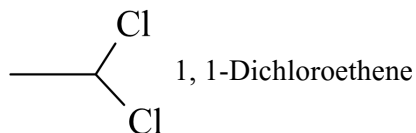
10. Ans (B)

Successive K_f decreases; $K_5 = K_1 \times K_2 \times K_3 \times K_4$

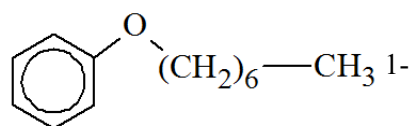
PART-2 : CHEMISTRY

SECTION-II

2. Ans (19.00)

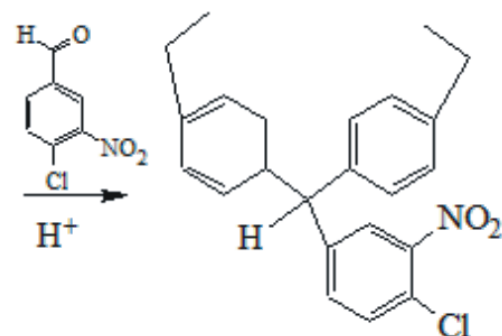
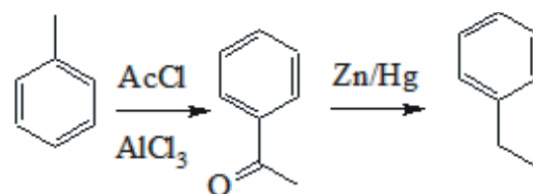
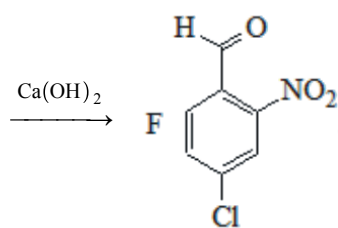
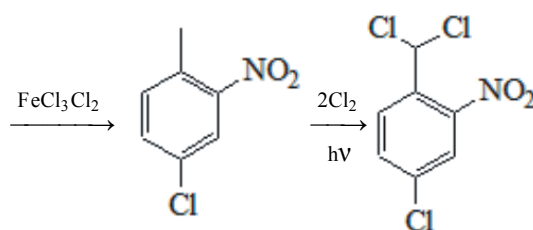
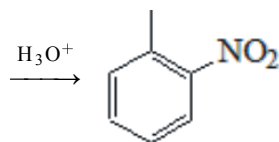
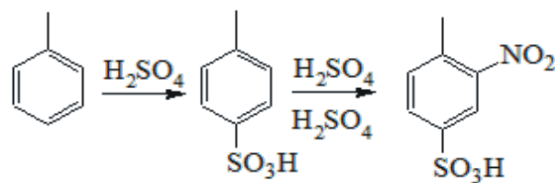


dimethylcyclohexane.

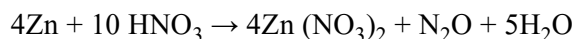


Phenoxyheptane

3. Ans (13.00)



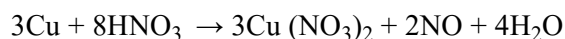
4. **Ans (47.62)**



0.1 mole

$$n\text{HNO}_3 = \frac{0.1}{4} \times 10 = \frac{1}{4} = 0.25 \text{ mole}$$

$$= 0.1 \times M \Rightarrow M = 2.5 \text{ M}$$



$$^n\text{Cu} = \frac{3}{8} n\text{HNO}_3$$

$$= \frac{3}{8} \times 2.5 \times \frac{80}{1000} = 75 \text{ mmol}$$

$$\therefore w\text{Cu} = (75 \times 10^{-3}) \times (63.5)$$

$$\Rightarrow \% \text{ Cu} = 47.625\%$$

5. **Ans (14.00)**

$$\pi = Pgh = CRT$$

$$\Rightarrow 1000 \times 10 \times \frac{h}{100} = c \times 8 \times 300$$

$$10^4 \times \frac{h}{100} = \left\{ \left[\frac{1 \times 10^{-3}}{1 + \left(\frac{h}{2} \times 100 \right) \times \frac{1}{1000}} \right] \times 1000 \right\} \times 8 \times 300$$

$$\Rightarrow h = \frac{480}{20 + h}$$

$$\Rightarrow h = 14 \text{ cm}$$

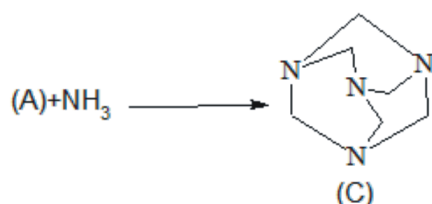
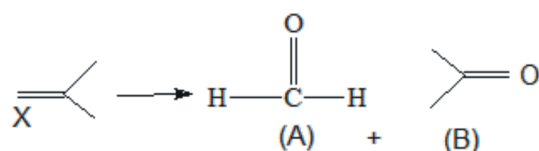
6. **Ans (6.00)**

a, d, e, f, g, h

7. **Ans (2.00)**

a, b

8. **Ans (8.00)**



PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (B,C)**

$$\text{As } SP = 6 = 3t^2 + 3 \Rightarrow t = 1$$

$$\Rightarrow P = (3, 6)$$

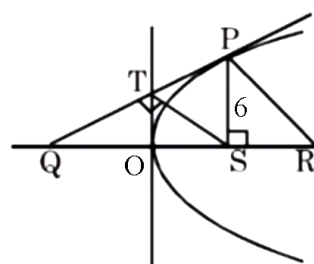
$$\text{Tangent at P : } 6y = 6(x + 3)$$

$$\Rightarrow x - y + 3 = 0$$

$$\Rightarrow T = (0, 3) \Rightarrow ST = 3\sqrt{2}$$

$$\text{Also } Q = (-3, 0) \Rightarrow QS = 6$$

$$\text{As given } SR = 6 \Rightarrow R(9, 0)$$



As we know if tangent and normal at any point P cuts axis at Q & R then $SP = SQ = SR$

According to figure PR is normal

Now,

$$\text{Area of PRST} = \text{Area of } \Delta PQR - \text{Area of } \Delta TQS$$

$$= \frac{1}{2} \times PS \times QR - \frac{1}{2} \times OT \times QS$$

$$= \frac{1}{2} \times 6 \times 12 - \frac{1}{2} \times 3 \times 6 = 27$$

Also as we know if tangents at P & M meet at point N, then

$$(\text{SN})^2 = SP \times SM$$

$$25 = 6 \times SM$$

$$\Rightarrow SM = \frac{25}{6} = 3t^2 + 3$$

$$\Rightarrow t^2 = \frac{7}{18} \Rightarrow t = \pm \frac{\sqrt{7}}{3\sqrt{2}}$$

Ordinate of M = 2at

$$= 2 \cdot 3 \cdot \left(\pm \frac{\sqrt{7}}{3\sqrt{2}} \right)$$

$$= \pm \sqrt{14}$$

$$\text{sum of squares} = 14 + 14 = 28$$

2. Ans (A,C)

Let $f(x) = ax^2 + bx + c$; $f(0) = c = 1$ using partial

fractions, let :

$$\frac{ax^2 + bx + 1}{x^2 \cdot (x+1)^3} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x+1} + \frac{D}{(x+1)^2} + \frac{E}{(x+1)^3}$$

here as integral $g(x)$ is a rational function $A = C = 0$:

$$\Rightarrow ax^2 + bx + 1 = B(x+1)^3 + Dx^2(x+1) + Ex^2$$

$$x = 0 \Rightarrow 1 = B$$

$$x = -1 \Rightarrow a - b + 1 = E$$

$$\text{coeff. of } x^3 \Rightarrow 0 = B + D \Rightarrow D = -1$$

$$\text{coeff. of } x^2 \Rightarrow a = 3B + D + E$$

$$a = 3 - 1 + E$$

$$a = 2 + a - b + 1$$

$$\Rightarrow b = 3$$

$$f(x) = 2ax + b \Rightarrow f(0) = b = 3$$

$$\text{taking : } \lim_{x \rightarrow \infty} (\sqrt{f(x)} - 1 - x) = \ell$$

$$\Rightarrow \lim_{x \rightarrow \infty} (\sqrt{ax^2 + 3x} - x) = \ell$$

$$\Rightarrow \lim_{x \rightarrow \infty} \frac{ax^2 + 3x - x^2}{\sqrt{ax^2 + 3x} + x} = \ell$$

here a must be $= 1$ for limit to exist.

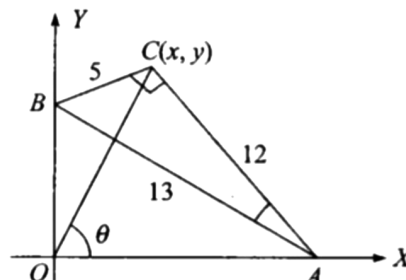
$$\text{Now, } \lim_{x \rightarrow \infty} \frac{3x}{\sqrt{x^2 + 3x} + x} = \frac{3}{1+1}$$

$$\Rightarrow \ell = \frac{3}{2}$$

3. Ans (B,C)

$$\text{As } \angle BCA + \angle BOA = 90^\circ + 90^\circ = 180^\circ$$

$\Rightarrow A, O, B, C$ are concyclic.



$$\Rightarrow \angle COB = \frac{\pi}{2} - \theta = \angle CAB$$

$$\tan\left(\frac{\pi}{2} - \theta\right) = \frac{5}{12} \Rightarrow \cot \theta = \frac{5}{12}$$

$$\text{Locus of } C(x, y) = \frac{y}{x} = \frac{12}{5}$$

$$\Rightarrow 12x = 5y$$

Also using Ptolemy's theorem :

$$BC \cdot OA + AC \cdot OB = AB \cdot OC$$

$$\Rightarrow 5 \cdot OA + 12 \cdot OB = 13 \cdot OC$$

$$\Rightarrow \frac{5}{13} OA + \frac{12}{13} OB = OC$$

$$\Rightarrow P = \frac{5}{13}, Q = \frac{12}{13}$$

4. Ans (A,B)

$$2x^2 - 2x + 1 = 0$$

$$m + n = 1, mn = \frac{1}{2}$$

$$\Rightarrow 2m = \frac{1}{n} \text{ or } \frac{1}{n^3} = 8m^3$$

$$g(x) = 8x^3 + \frac{1}{x^3}$$

Now,

$$g(m) + g(n) = 8m^3 + \frac{1}{m^3} + 8n^3 + \frac{1}{n^3}$$

$$= 8m^3 + 8n^3 + 8n^3 + 8m^3$$

$$= 16(m^3 + n^3) = -8$$

$$g(m) - g(n) = 0$$

$$\Rightarrow OQ = \sqrt{50} \cdot \cos \theta$$

$$\text{Area of } \triangle PQR = \frac{1}{2} \cdot OP \cdot OQ$$

$$= \sqrt{50} \cdot \sec \theta \cdot \sqrt{50} \cos \theta = \frac{50}{\sin \theta \cdot \cos \theta}$$

$$= \frac{100}{\sin 2\theta} \Rightarrow \min \text{ area} = 100$$

$$\text{when } \theta = 45^\circ$$

$$\Rightarrow K = \sqrt{50} \times \sec 45^\circ = \sqrt{50} \cdot \sqrt{2} = 10$$

$$(III) 2x^2 + xy + y^2 = 5 \rightarrow \text{center } (0, 0)$$

$$\text{put } x = r \cdot \cos \theta, y = r \sin \theta$$

$$\Rightarrow r^2 = \frac{5}{2\cos^2 \theta + \sin \theta \cdot \cos \theta + \sin^2 \theta}$$

$$= \frac{5}{2 + 2\cos^2 \theta + 2\sin \theta \cos \theta}$$

$$= \frac{5}{3 + \cos 2\theta + \sin 2\theta}$$

For length of major axis r is maximum

$$r_{\max}^2 = \frac{10}{3 - \sqrt{2}} \Rightarrow a = 3, b = \sqrt{2}$$

$$\Rightarrow a^2 \cdot b = 9 \cdot \sqrt{2} = \sqrt{162}$$

$$(IV) \text{ tangent to } \frac{x^2}{25} - \frac{y^2}{24} = 1 \text{ at } (5\sqrt{2}, 2\sqrt{6})$$

will be :

$$\frac{x \cdot 5\sqrt{2}}{25} - \frac{y \cdot 2\sqrt{6}}{24} = 1$$

intersecting tangent at vertices $x = \pm 5$ at

points A, B

$$\Rightarrow \pm \sqrt{2} - \frac{y}{2\sqrt{6}} = 1$$

$$\Rightarrow y = 2\sqrt{6}(\sqrt{2} - 1) \text{ or } -2\sqrt{6}(\sqrt{2} + 1)$$

Now segment of any tangent between tangent

at vertices subtends 90° at foci, circle drawn

through these 4 points will have A, B as end

points of diameter.

$$\Rightarrow K = \frac{2\sqrt{6}(\sqrt{2} - 1) - 2\sqrt{6}(\sqrt{2} + 1)}{2} = -2\sqrt{6}$$

$$\Rightarrow |2\sqrt{3} \cdot K| = 12 \cdot \sqrt{2}$$

8. Ans (D)

$$n(s) = {}^{30}C_4$$

$$(I) P = 1 - P(\text{are no are } \geq 15)$$

$$= 1 - \frac{{}^{16}C_4}{{}^{30}C_4} = \frac{731}{783}$$

$$(II) P = \frac{{}^{19}C_1 \times 1 \times {}^{10}C_2}{{}^{30}C_4} = \frac{19}{609}$$

$$(III) P = P(4 \text{ even}) + P((2E, 2O) + P(4O))$$

$$= \frac{{}^{15}C_4 + {}^{15}C_2 \times {}^{15}C_2 + {}^{15}C_4}{{}^{30}C_4} = \frac{2951}{5481}$$

$$(IV) P = 1 - P(\text{pqrs is odd})$$

$$= 1 - \frac{{}^{15}C_4}{{}^{30}C_4} = \frac{248}{261}$$

9. Ans (C)

$$(I) \int_0^1 (f(x) - f^{-1}(x)) dx = 0$$

$$(II) I = \frac{\int_0^{51} ([x] + 25) dx}{\int_0^{51} \{x\} dx}$$

$$= \frac{(0 + 1 + \dots + 50) + 25 \times 51}{51 \times \frac{1}{2}}$$

$$I = \frac{\frac{50 \times 51}{2} + 25 \times 51}{\frac{51}{2}} = 50 \times 2 = 100$$

$$(III) L = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \sin^2 \frac{r\pi}{2n}$$

$$L = \int_0^1 \sin^2 \frac{x\pi}{2} dx$$

$$L = \int_0^1 \sin^2 \left(\frac{\pi}{2} (1 - x) \right) dx$$

$$2L = \int_0^1 1 dx$$

$$L = \frac{1}{2}$$

$$(IV) J = \int_0^\infty x e^{-2x} dx$$

$$= \frac{x e^{-2x}}{-2} \Big|_0^\infty - \int_0^\infty \frac{e^{-2x}}{-2} dx$$

$$= (0 - 0) + \frac{1}{2} \frac{e^{-2x}}{-2} \Big|_0^\infty$$

$$J = \frac{1}{4}$$

$$\Rightarrow 8J = 2$$

10. Ans (B)

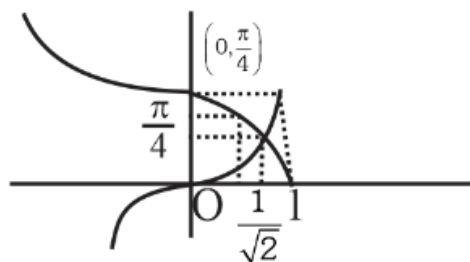
$$t_1 = (\sin^{-1} x)^{\sin^{-1} x}, t_2 = (\sin^{-1} x)^{\cos^{-1} x}$$

$$x \in (0, \cos 1)$$

$$1 < \cos^{-1} x < \frac{\pi}{2}$$

$$0 < \sin^{-1} x < \frac{\pi}{2} - 1$$

$$t_2 = (\sin^{-1} x)^{\cos^{-1} x}$$



(I) in this case

$$t_4 > t_3 > t_1 > t_2$$

$$(II) x \in \left(\cos 1, \frac{1}{\sqrt{2}} \right)$$

$$\frac{\pi}{4} < \cos^{-1} x < 1$$

$$\frac{\pi}{2} - 1 < \sin^{-1} x < \frac{\pi}{4}$$

$$\cos^{-1} x > \sin^{-1} x$$

$$(\cos^{-1} x)^{\cos^{-1} x} > (\sin^{-1} x)^{\cos^{-1} x}$$

$$t_4 > t_2 \quad \dots (i)$$

$$(\cos^{-1} x)^{\sin^{-1} x} > (\sin^{-1} x)^{\sin^{-1} x}$$

$$t_3 > t_1 \quad \dots \left(\frac{1}{2} \right)$$

$$t_3 > t_4$$

$$\text{So, } t_3 > t_4 > t_1 > t_2$$

$$(III) x \in \left(\frac{1}{\sqrt{2}}, \sin 1 \right)$$

$$\frac{\pi}{4} < \sin^{-1} x < 1, \quad \cos^{-1} x < \frac{\pi}{4}$$

$$\sin^{-1} x > \cos^{-1} x$$

$$t_2 > t_1 > t_4 > t_3$$

$$(IV) x \in (\sin 1, 1)$$

$$\sin^{-1} x > \cos^{-1} x$$

$$t_1 > t_2 > t_4 > t_3$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (3.00)

$$\log y = x \log \left(1 + \frac{1}{x} \right)$$

$$\frac{y_1(x)}{y} = \frac{x^2}{x+1} \left(-\frac{1}{x^2} \right) + \log \left(1 + \frac{1}{x} \right)$$

$$= -\frac{1}{x+1} + \log \left(1 + \frac{1}{x} \right) \dots (1)$$

$$y(2) = \frac{9}{4}$$

from equation (1),

$$y_1(x) = y \left(-\frac{1}{x+1} + \log \left(1 + \frac{1}{x} \right) \right)$$

$$y_2(x) = y_1 \left(-\frac{1}{x+1} + \log \left(1 + \frac{1}{x} \right) \right)$$

$$+ y(x) \left(\frac{1}{(x+1)^2} + \frac{x}{x+1} \left(-\frac{1}{x^2} \right) \right)$$

$$y_2(2) = y_1(2) \left(-\frac{1}{3} + \log \frac{3}{2} \right) + y(2) \left(\frac{1}{9} - \frac{1}{6} \right)$$

$$= \frac{9}{4} \left(-\frac{1}{3} + \log \frac{3}{2} \right)^2 - \frac{1}{8}$$

2. Ans (4.55 or 4.56)

Using cauchy - schwartz inequality :

$$2 \sin x \cdot \sin y + 3 \cos y + 6 \cos x \cdot \sin y$$

$$\leq \sqrt{2^2 + 3^2 + 6^2}$$

$$\times \sqrt{\sin^2 x \cdot \sin^2 y + \cos^2 y + \cos^2 x \cdot \sin^2 y}$$

$$\leq 7 \times 1$$

$$\Rightarrow \frac{\sin x \cdot \sin y}{2} = \frac{\cos y}{3} = \frac{\cos x \cdot \sin y}{6}$$

$$\text{using (i) \& (iii) } \tan x = \frac{1}{3}$$

$$\text{Using (i) \& (ii) } \sin x \cdot \tan y = \frac{2}{3}$$

$$\Rightarrow \tan y = \frac{2}{3} \times (\pm \sqrt{10})$$

$$\tan^2 x + \tan^2 y = \frac{41}{9}$$

3. Ans (5.00)

D.E. of given trajectory

$$(x^2 + y^2 - ay) = 0 \text{ is } 2x + 2y \frac{dy}{dx} = a \frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx} = \frac{2xy}{x^2 - y^2}$$

For orthogonal trajectory :

$$-\frac{dx}{dy} = \frac{2xy}{x^2 - y^2}$$

Put $y = vx$ and after solving it we get

$$x^2 + y^2 = kx \text{ which passes through } (4, 2)$$

$$\text{so } k = 5$$

$\therefore$ orthogonal trajectory will be

$$x^2 + y^2 - 5x = 0 \text{ which represents circle with diameter 5.}$$

4. Ans (85.00)

$$\text{Let } P(x) = x^2 + ax + b$$

Since $P(P(x)) = 0$ has roots 3, 4 so

Case-I :

$P(3) \neq P(4)$ and $P(3)$ and $P(4)$ will be roots of

$$P(x) = 0$$

$$\text{so } P(3) + P(4) = -a \text{ and}$$

$$P(3)P(4) = b$$

$$\Rightarrow a = \frac{-1}{2}, b = \frac{-21}{2}$$

$$\text{So, } (\alpha + \beta)^2 = 36,$$

Case-II :

$$P(3) = P(4) \Rightarrow \alpha + \beta = -7$$

$$\text{So, } (\alpha + \beta)^2 = 49$$

So reqd. sum = 85

5. Ans (2.80)

$$\alpha + \beta = -z_1, \quad \alpha\beta = z_2 + m$$

$$(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$$

$$= z_1^2 - 4(z_2 + m)$$

$$= z_1^2 - 4z_2 - 4m$$

$$= 16 + 20i - 4m$$

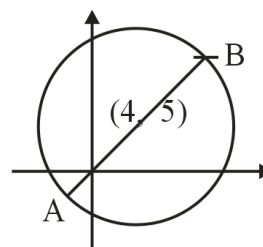
$$(\alpha - \beta)^2 = 4(4 + 5i - m)$$

$$|\alpha - \beta|^2 = 4|4 + 5i - m|$$

$$4 \times 7 = 4|4 + 5i - m|$$

$$|m - (4 + 5i)| = 7$$

So m lies on circle centre $(4, 5)$ and radius 7



$$|M|_{\max} = \sqrt{4^2 + 5^2} + 7 = 7 + \sqrt{41}$$

$$|M|_{\min} = \left| \sqrt{4^2 + 5^2} - 7 \right| = 7 - \sqrt{41}$$

their sum is

$$(7 + \sqrt{41}) + (7 - \sqrt{41})$$

$$\lambda = 14$$

$$\frac{\lambda}{5} = \frac{14}{5} = 2.80$$

6. Ans (2.50)

Coefficient of x^r in $(1 - x)^{-n}$ is ${}^{n+r-1}C_r$

$$\Rightarrow {}^{30}C_0 {}^{20}C_{10} + {}^{31}C_1 {}^{19}C_{10} +$$

$${}^{32}C_2 {}^{18}C_{10} + \dots + {}^{40}C_{10} {}^{10}C_{10} = {}^aC_b$$

Coefficient of x^{10} in $[(1 - x)^{-31} (1 - x)^{-11}]$

Coefficient of x^{10} in $(1 - x)^{-42}$

$$\Rightarrow {}^{42+10-1}C_{10} = {}^{51}C_{10} = {}^{51}C_{41}$$

$$a = 51, b = 41$$

$$a - b = 10 \Rightarrow \frac{a - b}{4} = \frac{10}{4} = 2.50$$

7. Ans (1.25)

$$\frac{\sum_{i=1}^{20} (x_i + i)}{20} = 15.5$$

$$\frac{\sum_{i=1}^{20} x_i}{20} + \frac{\sum_{i=1}^{20} i}{20} = 15.5$$

$$\bar{x} + \frac{20 \times 21}{2 \times 20} = 15.5$$

$$\bar{x} = 15.5 - 10.5 = 5 \quad \dots(1)$$

$$\sum_{i=1}^{20} (x_i - 5)^2 = 100$$

$$\sum x_i^2 + 25 \sum 1 - 10 \sum x_i = 100$$

$$\sum x_i^2 + 25 \times 20 - 10 \times 5 \times 20 = 100$$

$$\sum x_i^2 = 1000 + 1000 - 500 = 600$$

$$SD = \sqrt{\frac{\sum x_i^2}{20} - (\bar{x})^2} = \sqrt{\frac{600}{20} - (5)^2}$$

$$= \sqrt{30 - 25} = \sqrt{5}$$

$$\Rightarrow \text{var}(x_i - 2) = \text{var}(x_i) = 5$$

8. Ans (2.50)

$$\text{adj}A = \begin{bmatrix} 2 & 0 & 0 \\ -1 & 2 & 0 \\ 3 & 1 & 4 \end{bmatrix}, |A| > 0$$

$$|\text{adj}A| = |A|^2 = 16$$

$$\text{or } |A| = 4$$

$$\therefore \text{adj}(\text{adj}A) = |A|.A = 4A$$

then diagonal elements of A

$$A = \frac{1}{4} \begin{bmatrix} 8 & & \\ & 8 & \\ & & 4 \end{bmatrix} \Rightarrow \frac{1}{2} \text{tr}(A) = \frac{5}{2} = 2.50$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

FULL SYLLABUS

12-02-2023

JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-II)

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	B	C	A	B				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	A,C	A,B,C,D	A,B	B,C	A,C,D	A,D		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	2	8	5	1	5	5	5	9

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4				
	A.	B	B	A	B				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	A,C,D	A,B,C	A,D	B,C,D	A,D	B,C		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	5	4	6	6	8	8	3	5

PART-3 : MATHEMATICS

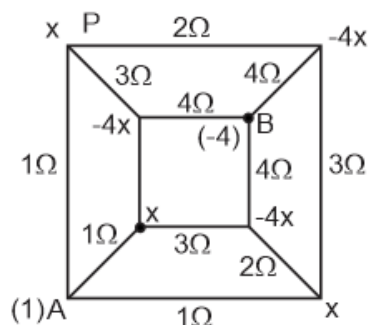
SECTION-I (i)	Q.	1	2	3	4				
	A.	A	C	B	A				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	A,C	A,D	B,C	A,B,C,D	A,B,C,D	A,B,D		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	8	1	1	7	2	5	2	3

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (B)



Applying nodal analysis at point p

$$\frac{x + 4x}{3} + \frac{x + 4x}{2} + \frac{x - 1}{1} = 0$$

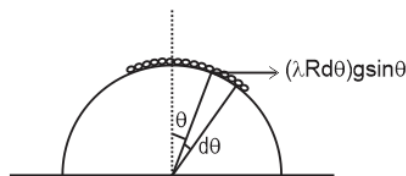
$$10x + 15x + 6x - 6 = 0$$

$$x = \frac{6}{31}$$

$$R_{\text{net}} = \frac{V_{\text{net}}}{i_{\text{net}}} = \frac{5}{3(1-x)} = \frac{5}{3 \times \frac{25}{31}} = \frac{31}{15}$$

$$\text{Ans. } \frac{31}{15} \Omega$$

2. Ans (C)



Net tangential force

$$= \lambda Rg \int_0^{\alpha_2} \sin \theta d\theta - \lambda Rg \int_0^{\alpha_1} \sin \theta d\theta$$

$$F = \lambda Rg [\cos \alpha_1 - \cos \alpha_2] = \lambda R(\alpha_1 + \alpha_2)a$$

$$\frac{\lambda Rg(\cos \alpha_1 - \cos \alpha_2)}{\lambda R(\alpha_1 + \alpha_2)} = a$$

$$\frac{g(\cos \alpha_1 - \cos \alpha_2)}{(\alpha_1 + \alpha_2)} = a$$

3. Ans (A)

Due to Heisenberg's uncertainty initial velocity of particle can't be ideally horizontal.

So, particles will be having some initial velocity in perpendicular direction also.

$$\Delta p_y \Delta y \simeq \hbar$$

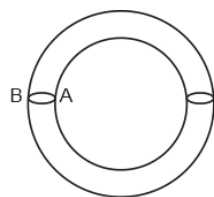
$$v_y \simeq \frac{\hbar}{mD_0} \quad (\because \Delta y \simeq D_0)$$

$$T = \frac{L}{v}$$

$$\text{So, } \Delta D = 2v_y T = 2 \frac{\hbar}{mD_0} \frac{L}{v}$$

$$D_L = D_0 + 2 \frac{\hbar L}{mD_0 v}$$

4. Ans (B)



Pressure at point just inside point B

$$P_B = \rho_0 + \frac{T}{20} + \frac{T}{\rho_B}$$

Pressure at point just inside point A

$$P_A = P_0 - \frac{T}{10} + \frac{T}{5}$$

$$P_B = P_A$$

$$\frac{1}{20} + \frac{1}{\rho_B} = \frac{-1}{10} + \frac{1}{5}$$

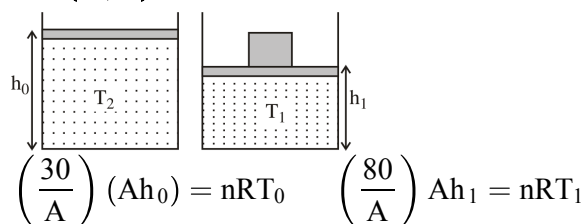
$$\frac{1}{\rho_B} = \frac{1}{5} - \frac{1}{10} - \frac{1}{20} = \frac{4-2-1}{20} = \frac{1}{20}$$

$$\rho_B = 20 \text{ cm}$$

PART-1 : PHYSICS

SECTION-I (ii)

5. Ans (A,C)



$$\left(\frac{30}{A}\right)(Ah_0) = nRT_0 \quad \left(\frac{80}{A}\right)Ah_1 = nRT_1$$

$$h_0 = \frac{nRT_0}{30} \quad h_1 = \frac{nRT_1}{80}$$

$$w_{gr} = \Delta U$$

$$80 \times (h_0 - h_1) = n \times \frac{3}{2} R(T_1 - T_0)$$

$$80 \times \left(\frac{nRT_0}{30} - \frac{nRT_1}{80}\right) = n \frac{3}{2} R(T_1 - T_0)$$

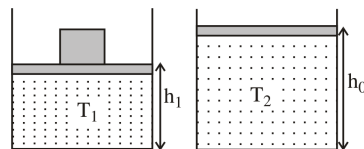
$$\frac{8}{3} T_0 - T_1 = \frac{3}{2} T_1 - \frac{3}{2} T_0$$

$$\left(\frac{8}{3} + \frac{3}{2}\right) T_0 = \frac{5}{2} T_1$$

$$\left(\frac{16+9}{6}\right) T_0 = \frac{5}{2} T_1$$

$$\frac{25}{6} T_0 = \frac{5}{2} T_1$$

$$T_1 = \frac{2}{5} \times \frac{25}{6} T_0 = \frac{5}{3} T_0$$



$$\frac{(80)}{A}(Ah_1) = nRT_1 \quad \frac{(30)}{A}(Ah_0) = nRT_2$$

$$h_1 = \frac{nRT_1}{80} \quad h_0 = \frac{nRT_2}{30}$$

$$w_{qr} = \Delta U$$

$$-30 \times (h_0 - h_1) = n \times \frac{3}{2} R(T_2 - T_1)$$

$$-30nr \left[\frac{T_2}{30} - \frac{T_1}{80}\right] = nR \times \frac{3}{2} [T_2 - T_1]$$

$$-T_q + \frac{3}{8} T_1 = \frac{3}{2} T_2 - \frac{3}{2} T_1$$

$$\left(\frac{3}{8} + \frac{3}{2}\right) T_1 = \left(\frac{3}{2} + 1\right) T_2$$

$$\left(\frac{3+12}{8}\right) T_1 = \frac{5}{2} T_2$$

$$\frac{2}{5} \times \frac{15}{8} T_1 = T_2$$

$$\frac{3}{4} \times \frac{5}{3} T_0 = T_2$$

$$\frac{5}{4} T_0 = T_2$$

6. Ans (A,B,C,D)

$$\text{Volume of tetrahedron} = \frac{a}{6\sqrt{2}}$$

(C) Net force by liquid = Buoyant force =

$$\frac{a^3}{6\sqrt{2}}\rho g = B$$

$$(B) \text{ Force on the bottom} = h\rho g \times \frac{\sqrt{3}}{4}a^2 = F_1$$

(A) Net force on all the side faces =

$$F_1 - B = \frac{\sqrt{3}\rho g h a^2}{4} - \frac{a^3\rho g\sqrt{2}}{12}$$

(D) Vertical force of each face is $\frac{1}{3}$ of F_{net}

vertical force as in option A.

The net force on each face = $\frac{F_0}{3 \cos \theta}$, where θ

is angle the net force makes with vertical

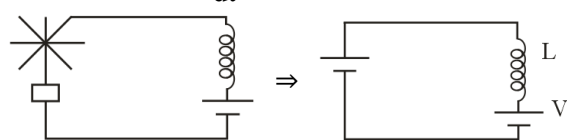
$$\therefore \cos \theta = \frac{1}{3}$$

$$f = F_0$$

7. Ans (A,B)

$$V - \frac{B\omega R^2}{2} - \frac{Ldi}{dt} = 0$$

$$-\frac{BR^2}{2} \frac{d\omega}{dt} = L \frac{d^2i}{dt^2} \quad \dots(1)$$



$$dF = i dx B$$

$$d\tau = x dF = i B x dx$$

$$\tau = iB \int_0^R x dx$$

$$\tau = \frac{iBR^2}{2} = I\alpha = I \frac{d\omega}{dt}$$

$$\frac{d\omega}{dt} = \frac{iBR^2}{2I}$$

Putting in equation (1)

$$-i \frac{B^2 R^4}{4I} = L \frac{d^2i}{dt^2}$$

$$-\frac{B^2 R^4}{4IL} i = \frac{d^2i}{dt^2}$$

$$\text{So, } i = i_0 \sin\left(\frac{BR^2}{2\sqrt{IL}}t + \phi\right)$$

$$\text{At } t = 0, i = \omega = 0$$

$$\frac{di}{dt} = \frac{V}{L}$$

$$\text{So, } \phi = 0$$

$$i = i_0 \sin\left(\frac{BR^2}{2\sqrt{IL}}t\right)$$

$$\frac{di}{dt} = i_0 \frac{BR^2}{2\sqrt{IL}} \cos\left(\frac{BR^2}{2\sqrt{IL}}t\right)$$

$$\frac{V}{L} = i_0 \frac{BR^2}{2\sqrt{IL}}$$

$$i_0 = \frac{2V}{BR^2} \sqrt{\frac{I}{L}}$$

$$\text{So, } i = \frac{2V}{BR^2} \sqrt{\frac{I}{L}} \sin\left(\frac{BR^2}{2\sqrt{IL}}t\right)$$

$$i \text{ is max when } \theta = \frac{\pi}{2}$$

$$\frac{BR^2}{2\sqrt{IL}}t = \frac{\pi}{2}$$

$$t = \frac{\pi\sqrt{IL}}{BR^2}$$

$$\frac{d\omega}{dt} = \frac{iBR^2}{2I} = i_0 \sin(at)$$

$$(\text{where } i_0 = \frac{2V}{BR^2} \sqrt{\frac{I}{L}}, a = \frac{BR^2}{\sqrt{2IL}})$$

$$\omega = f(t)$$

$$KE = \frac{1}{2}I\omega^2$$

8. Ans (B,C)

$$E_{\text{circular}} = E_{\text{ellipse}}$$

$$-\frac{GMm}{2R} = -\frac{GMm}{2a}$$

$$a = R = 40 \times 10^7 \text{ m}$$

$$OB = a(1 - e) = 8 \times 10^7$$

$$1 - e = \frac{1}{5}$$

$$e = \frac{4}{5} = 0.8$$

$$v_A = \sqrt{\frac{GM}{R}}$$

$$-\frac{GMm}{2R} = \frac{1}{2}mv_B^2 - \frac{GMm}{(OB)}$$

$$v_B = \sqrt{\left(\frac{GM}{8 \times 10^7} - \frac{GM}{2 \times 40 \times 10^7}\right) \times 2}$$

$$\frac{v_B}{v_A} = \sqrt{\left(\frac{\frac{1}{8} - \frac{1}{80}}{\frac{1}{40}}\right) \times 2} = \sqrt{\frac{9 \times 40 \times 2}{80 \times 1}} = 3$$

9. Ans (A,C,D)

$$\vec{E} = (3\hat{j} + b\hat{k}) \times 10^{-3} \sin[10^7(x + 2y + 3z - \beta t)]$$

$$x + 2y + 3z - \beta t = (\hat{i} + 2\hat{j} + 3\hat{k}) \cdot (\hat{i} + 2\hat{j} + 3\hat{k}) - \beta t$$

So, $\hat{i} + 2\hat{j} + 3\hat{k}$ is direction of propagation of wave.

$\vec{E} \perp$ dir. of propagation

$$(3\hat{j} + b\hat{k}) \cdot (\hat{i} + 2\hat{j} + 3\hat{k}) = 0$$

$$6 + 3b = 0$$

$$b = -2$$

$$\omega = 10^7 \beta$$

$$k = 10^7 \times \sqrt{1^2 + 2^2 + 3^2} = 10^7 \sqrt{14}$$

$$c = \frac{\omega}{k}$$

$$\beta = 3\sqrt{14} \times 10^8$$

$$E_0 = (\sqrt{3^2 + b^2}) \times 10^{-3} = \sqrt{13} \times 10^{-3}$$

$$\text{Avg. energy density} = \frac{1}{2} \epsilon_0 E_0^2$$

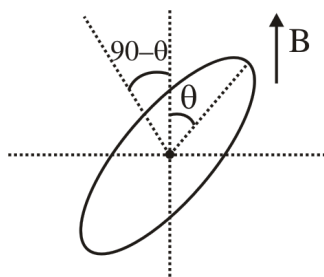
$$= 6.5 \times 10^{-6} \epsilon_0 \text{ SI unit}$$

10. Ans (A,D)

Let area is A

& Resistance R

$$\phi = BA \sin \theta$$



Due to change in flux, current induces

$$E = \frac{d\phi}{dt} = BA \cos \theta \frac{d\theta}{dt}$$

$$I = \frac{E}{R} = \frac{BA \cos \theta}{R} \frac{d\theta}{dt}$$

Now current carrying loop in magnetic field experience torque

$$\vec{\tau} = \vec{M} \times \vec{B} = IAB \sin(90^\circ - \theta)$$

$$\tau = IAB \cos \theta$$

$$\tau = \frac{B^2 A^2 \cos^2 \theta}{R} \frac{d\theta}{dt}$$

$$\tau_{\text{net}} = \tau_{\text{mag}} + \tau_{\text{tor}}$$

$$I \frac{d^2 \theta}{dt^2} = - \frac{B^2 A^2 \cos^2 \theta}{R} \frac{d\theta}{dt} - k\theta$$

θ is small, so $\cos \theta \rightarrow 1$

$$\frac{I d^2 \theta}{dt^2} = - \frac{B^2 A^2}{R} \frac{d\theta}{dt} - k\theta$$

Putting the value

$$0.2 \frac{d^2 \theta}{dt^2} = - \frac{10^{-4} \times 10^{-2}}{10^{-6}} \frac{d\theta}{dt} - 3.7\theta$$

$$0.2 \frac{d^2 \theta}{dt^2} + \frac{d\theta}{dt} + 3.7\theta = 0$$

$$\frac{d^2 \theta}{dt^2} + 5 \frac{d\theta}{dt} + 18.5\theta = 0$$

Auxiliary equation

$$m^2 + 5m + 18.5 = 0$$

$$m = \frac{-5 \pm \sqrt{25 - 74}}{2} = \frac{-5 \pm 7i}{2}$$

$$= \frac{-5}{2} \pm \frac{7}{2}i$$

$$\text{So, } \theta = e^{-\frac{5}{2}t} \left(A \cos \frac{7}{2}t + B \sin \frac{7}{2}t \right)$$

$$\text{At } t = 0 \quad \theta = \delta$$

$$\text{So, } \delta = A$$

$$\text{At } t = 0 \quad \frac{d\theta}{dt} = 0$$

$$\frac{d\theta}{dt} = e^{-\frac{5}{2}t} \left[-A \frac{7}{2} \sin \frac{7}{2}t + B \frac{7}{2} \cos \frac{7}{2}t \right] + \left(A \cos \frac{7}{2}t + B \sin \frac{7}{2}t \right) \left(-\frac{5}{2} \right) e^{-\frac{5}{2}t}$$

Putting $t = 0$,

$$0 = B \frac{7}{2} - \frac{5}{2}A$$

$$\text{So, } B = \frac{5}{7} \delta$$

$$\theta = e^{\frac{5}{2}t} \left[\delta \cos \frac{7}{2}t + \frac{5}{7} \delta \sin \frac{7}{2}t \right]$$

$$\theta = \delta e^{-\frac{5}{2}t} \left[\cos \frac{7}{2}t + \frac{5}{7} \sin \frac{7}{2}t \right]$$

$$\text{So, } \gamma = \frac{7}{2} \quad \frac{\beta}{\frac{7}{2}} = \frac{5}{7}$$

$$\beta = \frac{5}{7} \times \frac{7}{2} = \frac{5}{2}$$

PART-1 : PHYSICS

SECTION-III

1. **Ans (2)**

Time taken depends inversely on $\sqrt{a}$

(a = acceleration)

a for q = a_0

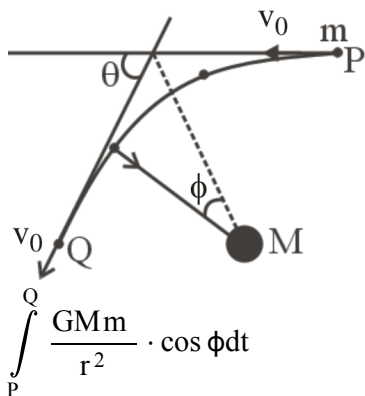
a for 4q = $4a_0$

time for 4q is $\frac{t_0}{2}$

2. **Ans (8)**

$$|\Delta \vec{P}| = 2mv_0 \sin \frac{\theta}{2}$$

$$|\Delta \vec{P}| = \left| \int \vec{F} dt \right|$$



$$\int_P^Q \frac{GMm}{r^2} \cdot \cos \phi dt$$

But $mr^2\omega = mv_0b$ (in L)

$$r^2 = \frac{v_0b}{d\phi} dt$$

$$\int_P^Q \frac{GMm}{r^2} \cos \phi dt = \int_{-(\frac{\pi}{2} + \frac{\theta}{2})}^{(\frac{\pi}{2} + \frac{\theta}{2})} \frac{GMm}{v_0b} \cos \phi d\phi$$

$$= \frac{GMm}{v_0b} 2 \cos \frac{\theta}{2}$$

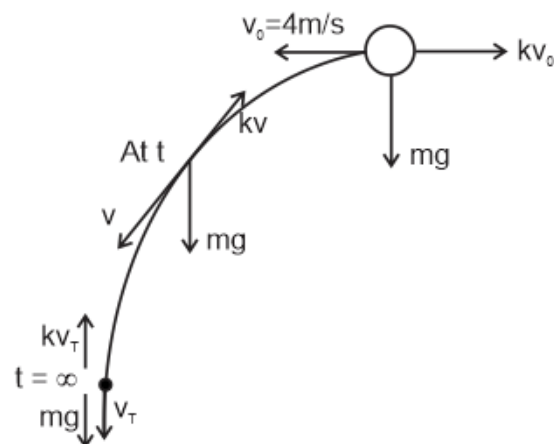
$$\text{So, } \frac{GMm}{v_0b} 2 \cos \frac{\theta}{2} = 2mv_0 \sin \frac{\theta}{2}$$

$$\tan \frac{\theta}{2} = \frac{GM}{v_0^2b}$$

$$\Rightarrow \theta = 2 \tan^{-1} \left(\frac{GM}{v_0^2b} \right) = 2 \tan^{-1} \left(\frac{GM}{4b} \right)$$

Ans. : $2 \times 4 = 8$

3. **Ans (5)**



With respect to wind blowing towards right. Object will have velocity 4 m/s towards left at $t = 0$.

So at $t \rightarrow \infty$ $kv_T = mg$

$$v_T = \frac{mg}{k} = 3 \text{ m/s}$$

with respect to ground.

$$v = \sqrt{3^2 + 4^2} = 5 \text{ m/s}$$

4. **Ans (1)**

Analog to gauss law

$$\text{Edr} = \text{mass} \times 4\pi G \quad G \rightarrow \frac{1}{4\pi\epsilon_0}$$

$$\oint E ds = \int_0^r \rho(r) 2\pi r dr L \times 4\pi G$$

$$E \cdot 2\pi r L = \int_0^r \rho(r) 2\pi r dr L \times 4\pi G$$

Diff both side

$$E 2\pi L \frac{dr}{dr} = \rho(r) 2\pi r L 4\pi G \frac{dr}{dr} - 0$$

$$E = \rho(r) r 4\pi G$$

$$F_{\text{centripetal}} = dm\omega^2 r$$

$$dmE = dm\omega^2 r$$

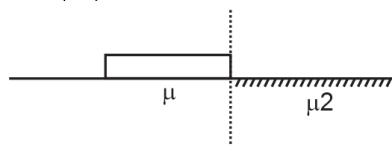
$$E = \omega^2 r$$

$$\rho(r) r 4\pi G = \omega^2 r$$

$$\rho(r) = \frac{\omega^2}{4\pi G} = \text{constant}$$

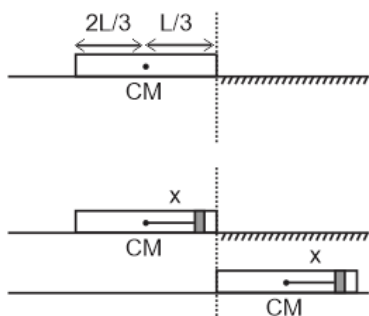
$$\text{So, } \frac{\rho \left(\frac{3R}{4} \right)}{\rho \left(\frac{R}{2} \right)} = 1$$

5. Ans (5)



Since $\lambda = \lambda_0 x$ center of mass will be at distance $\frac{2L}{3}$ from left end.

$$x_{cm} = \frac{\int x dm}{\int dm} = \frac{x_0 \int_0^L x^2 dx}{x_0 \int_0^L x dx} = \frac{2}{3}L$$



$$dw = \mu dm g \left(\frac{L}{3} - x \right) + 2\mu dm g \left(\frac{2L}{3} + x \right)$$

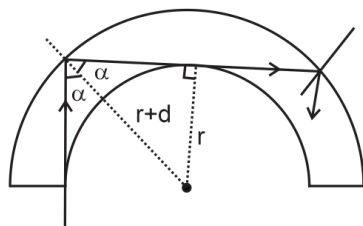
$$dw = \mu dm g \left(\frac{L}{3} + \frac{4L}{3} \right) + \mu dm g (2x - x)$$

$$w = \mu g \frac{5L}{3} \int dm + \mu g \int dm x$$

$$w = \frac{\mu g 5L}{3} M$$

Ans. 5

6. Ans (5)



$$\sin \alpha = \frac{r}{r+d}$$

For TIR

$$\sin \alpha \geq \frac{1}{\mu}$$

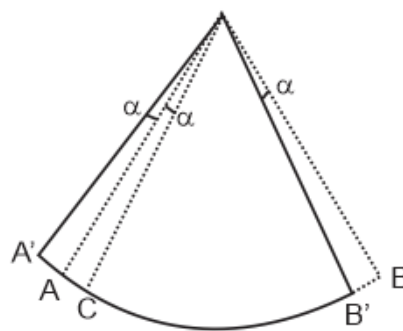
$$\frac{10}{10+d} \geq \frac{1}{1.5}$$

$$15 \geq 10+d$$

$$5 \geq d$$

$$d_{\max} = 5 \text{ cm}$$

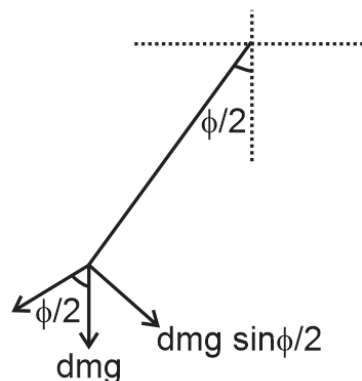
7. Ans (5)



After small angular displacement we will calculate torque. Torque due to B'C is zero. So, torque due to A'C is torque of oscillation. A'C is small, so we can consider it as point object at A.

$$dm = \lambda R 2\alpha$$

$$dm = \frac{m}{R\phi} R 2\alpha = \frac{2m\alpha}{\phi}$$



$$\tau = (dm)g \sin \frac{\phi}{2} R$$

$$\tau = \frac{2m\alpha}{\phi} g \sin \frac{\phi}{2} R$$

$$\tau = C\alpha$$

$$T = 2\pi \sqrt{\frac{I}{C}}$$

$$I = mR^2$$

$$T = 2\pi \sqrt{\frac{R\phi}{2g \sin(\frac{\phi}{2})}}$$

$$C = \frac{2mgR \sin \frac{\phi}{2}}{\phi}$$

$$T = 2\pi \sqrt{\frac{R\phi}{2g \sin(\frac{\phi}{2})}}$$

$$\text{If } R = \frac{3}{10\pi} m, \phi = \frac{\pi}{3}$$

$$T = \frac{\pi}{5}$$

8. **Ans (9)**

$$\frac{3}{4}\lambda = \ell\lambda = \frac{4\ell}{3} = \frac{v}{f}$$

$$\Rightarrow \lambda' = \frac{v}{\left(\frac{32f}{15}\right)} = \frac{15}{32} \left(\frac{4\ell}{3}\right) = \frac{5\ell}{8}$$

Possible positions are $\frac{\lambda'}{4}, \frac{3\lambda'}{4}, \frac{5\lambda'}{4}, \dots$

$$= \frac{5\ell}{32}, \frac{15\ell}{32}, \frac{25\ell}{32}, \frac{35\ell}{32}, \frac{45\ell}{32}$$

Shift wrt initial length for above

$$\Delta L_1 = \ell - \frac{5\ell}{32} = \frac{27\ell}{32} \quad \Delta L_2 = \frac{17\ell}{32}$$

$$\Delta L_3 = \frac{4\ell}{32} \quad \Delta L_4 = \frac{3\ell}{32}$$

$$\Delta L_5 = \frac{13\ell}{32}$$

From above minimum shift is

$$\Delta L_4 = \frac{3\ell}{32} = \frac{3}{32} \times 96 = 9$$

PART-2 : CHEMISTRY

SECTION-I (i)

1. **Ans (B)**

Distance between the planes OPQ and

$$ABG = 2R\sqrt{\frac{2}{3}}$$

$$\text{Distance of THV from O} = \sqrt{\frac{3}{2}}R$$

$$\therefore x = \left[2R\sqrt{\frac{2}{3}} - \sqrt{\frac{3}{2}}R \right]$$

$$\& y = R\sqrt{\frac{2}{3}}$$

$$x < y$$

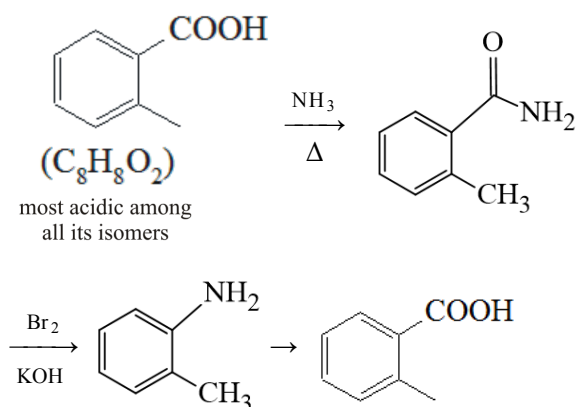
2. **Ans (B)**

PbSO₄(s) : White and CuSO₄(aq) : Blue

3. **Ans (A)**

Theory

4. **Ans (B)**



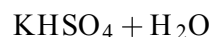
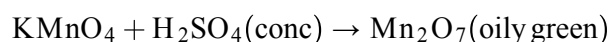
PART-2 : CHEMISTRY

SECTION-I (ii)

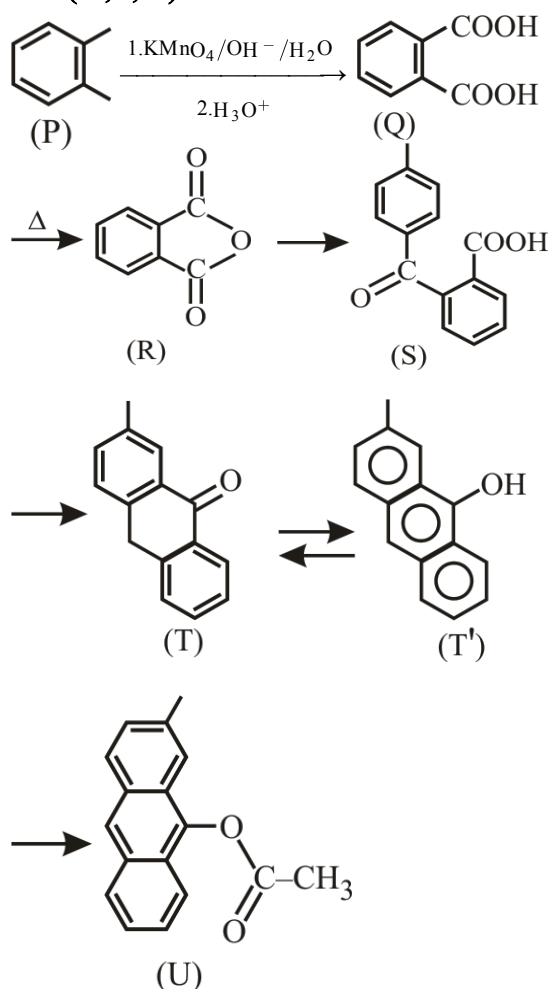
5. **Ans (A,C,D)**

NCERT THEORY

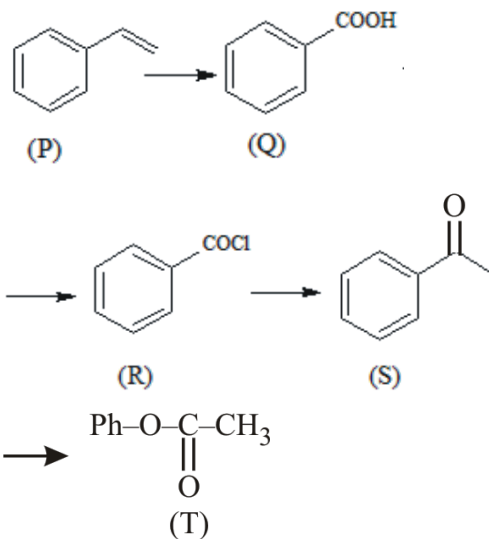
7. **Ans (A,D)**



8. **Ans (B,C,D)**



9. Ans (A,D)



10. Ans (B,C)

$$\rho_{\text{calculated}} = \frac{PM}{RT} = \frac{8 \times 64}{0.08 \times 400} = 16$$

$$\rho_{\text{real}} = 10 \text{ g/l}$$

$$\therefore Z_r = \frac{V_r}{V_i} = \frac{\rho_i}{\rho_r} = \frac{16}{10} = 1.6$$

PART-2 : CHEMISTRY

SECTION-III

1. Ans (5)

$$E^\circ_{\text{cell}} = +0.3V$$

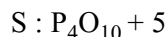
$$\therefore \text{Energy incident} = 0.3 + 5.4 = 5.7 \text{ eV}$$

$$\text{Now incident energy} = 2 \times 5.7 = 11.4$$

$$\therefore K.E_{\text{max}} = 11.4 - 5.4 = 6.0 \text{ eV}$$

$$\lambda_d \simeq \sqrt{\frac{150}{V}} = 5\text{\AA}$$

3. Ans (6)



4. Ans (6)

Volume of one cube = V of 10^{12} smaller cubes

$$\Rightarrow (1 \text{ cm})^3 = 10^{12} (a^3)$$

$$\text{Total area} = 10^{12} \times 6 \times a^2$$

$$= 6 \text{ m}^2$$

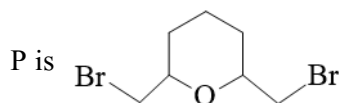
5. Ans (8)

$$\Delta S - 1 \times 8 \times \ln \frac{6}{2} = 8$$

6. Ans (8)

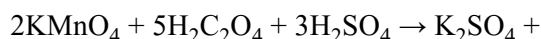
$$\text{COD} = \frac{0.1 \times (21 - 20) \times 8 \times 1000}{100} = 8$$

7. Ans (3)



stereoisomer are 3.

8. Ans (5)



$$nf(\text{H}_2\text{SO}_4) = \frac{10}{3} = \frac{2}{3}x \Rightarrow x = 5$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A)

$$\int (\sin(n+1)x + x) \sin^n x \, dx$$

$$\int \sin(n+1)x \cdot \sin^n x \cos x \, dx +$$

$$\int \cos(n+1)x \cdot (\sin x)^{(n+1)} \, dx$$

$$\frac{(\sin x)^{(n+1)}}{(n+1)} \sin(n+1)x - \int \cos(n+1)x \cdot (\sin x)^{n+1} \, dx$$

$$\int \cos((n+1)x) (\sin x)^{(n+1)} \, dx$$

$$\frac{(\sin x)^{(n+1)}}{(n+1)} \sin(n+1)x + c$$

$$F_n(\pi) = 0 \Rightarrow c = 0$$

$$\text{Hence } F_n(x) = \frac{(\sin x)^{(n+1)}}{(n+1)} \sin(n+1)x$$

$$F_{16}\left(\frac{3\pi}{2}\right) = -\frac{1}{17} \sin \frac{51\pi}{2} = a$$

$$a = \frac{1}{17}$$

2. Ans (C)

Here $f(x) = -x^2 + px - 1 - (p+1)^2$

$$f(x) = -2x + p = 0 \Rightarrow x = \frac{p}{2}$$

Case-1 :

If $p \leq 0$, then maximum value $f(x)$ is at $x = 0$

$$f(0) = -1 - (p+1)^2 = -5$$

$$(p+1)^2 = 4$$

$$1 + p = 2, -2,$$

$$\Rightarrow p = -3 \text{ or } p = 1$$

$$p = -3, (p = 1 \text{ not possible})$$

Case-2 :

$0 < p < 2$ then maximum value at $x = \frac{p}{2}$

$$f\left(\frac{p}{2}\right) = -5$$

$$\Rightarrow -\frac{p^2}{4} + \frac{p^2}{2} - 1 - (p+1)^2 = -5$$

$$\Rightarrow 3p^2 + 8p - 12 = 0$$

$$\Rightarrow p = \frac{-4 \pm 2\sqrt{13}}{3}$$

$$p = \frac{-4 + 2\sqrt{13}}{3} \text{ is possible}$$

$$[p] = 1$$

Case-3

For $p \geq 2$ then maximum value at $x = 1$.

$$f(1) = -5 \Rightarrow -1 + p - 1 - (p+1)^2 = -5$$

$$(p+1)^2 + 2 - p = 5$$

$$p^2 + p - 2 = 0$$

$$(p+2)(p-1) = 0$$

$$p = -2 \text{ \& } p = 1 \text{ both are not possible}$$

So possible values of p in above three cases are

$$p = -3, p = \frac{-4 + 2\sqrt{13}}{3}$$

3. Ans (B)

Let r be an integer such that $p(r) = 0$

$$P(x) = (x-2)q(x) + P(2)$$

$$P(r) = (r-2)q(r) + 13$$

$$q(r) = \frac{-13}{r-2}$$

$$\text{Now, } r-2 = \pm 1, \pm 13$$

$$r = 3, 1, 15, -11$$

$$P(x) = (x-10)q(x) + P(10)$$

$$P(r) = 0 \Rightarrow q(r) = \frac{-5}{r-10}$$

$$r-10 = 1, -1, 5, -5$$

$$r = 11, 9, 15, 5$$

$$\text{Thus } r = 15.$$

4. Ans (A)

Let the equation of plane passing through

origin is $ax + by + cz = 0$

$$\frac{|2a - 2b - c|}{\sqrt{2^2 + 2^2 + 1} \sqrt{a^2 + b^2 + c^2}} = \frac{|8a + b - 4c|}{\sqrt{8^2 + 1^2 + 4^2} \sqrt{a^2 + b^2 + c^2}}$$

$$\frac{|2a - 2b - c|}{3} = \frac{|8a + b - 4c|}{9}$$

$$3(2a - 2b - c) = 8a + b - 4c$$

$$6a - 6b - 3c = 8a + b - 4c$$

$$2a + 7b - c = 0$$

or

$$3(2a - 2b - c) = -8a - b + 4c$$

$$6a - 6b - 3c = -8a - b + 4c$$

$$14a - 5b - 7c = 0$$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. Ans (A,C)

$$z = \frac{(z_1 + z_2)(z_2 + z_3)(z_3 + z_4) \dots (z_{10} + z_{11})(z_{11} + z_1)}{z_1 z_2 z_3 \dots z_{11}}$$

$$z^2 = \frac{(z_1 + z_2)^2(z_2 + z_3)^2(z_3 + z_4)^2 \dots (z_{10} + z_{11})^2(z_{11} + z_1)^2}{z_1^2 z_2^2 z_3^2 \dots z_{11}^2}$$

$$z^2 = \frac{(z_1^2 + z_2^2 + 2z_1 z_2)(z_2^2 + z_3^2 + 2z_2 z_3) \dots (z_{11}^2 + z_1^2 + 2z_{11} z_1)}{z_1^2 z_2^2 z_3^2 \dots z_{11}^2}$$

$$z^2 = \frac{[z_1^2 + z_2^2 + 2z_1 z_2]}{z_1 z_2} \frac{[z_2^2 + z_3^2 + 2z_2 z_3]}{z_2 z_3} \dots \frac{[z_{11}^2 + z_1^2 + 2z_{11} z_1]}{z_{11} z_1}$$

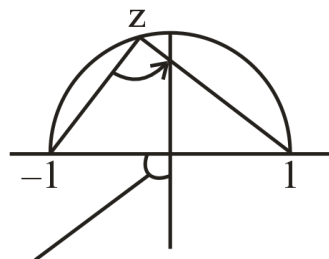
$$z^2 = \left[\frac{z_1}{z_2} + \frac{z_2}{z_1} + 2 \right] \left[\frac{z_2}{z_3} + \frac{z_3}{z_2} + 2 \right] \dots \left[\frac{z_{11}}{z_1} + \frac{z_1}{z_{11}} + 2 \right]$$

$$\frac{z_1}{z_2} + \frac{z_2}{z_1} = \text{purely real}$$

$z = \text{purely real}$

$$i^i = \left(e^{i\frac{\pi}{2}} \right)^i = e^{-\frac{\pi}{2}} = \text{purely real}$$

$$\arg z = -\frac{2\pi}{3} \quad \arg \left(\frac{z-1}{z+1} \right) = \frac{\pi}{4}$$



Total number of solution = zero.

6. Ans (A,D)

Differential equation can be written as :

$$(p - x)(p - 2 \sin x)(2p + \cos x) = 0$$

Which has solution :

$$\left(y - \frac{x^2}{2} - c \right) \cdot (y + 2 \cos x - k) \cdot (2y + \sin x - \lambda) = 0$$

7. Ans (B,C)

$$q = \sum_{r=1}^{\infty} \frac{1}{2^r} \left(1 - \frac{1}{2^r} \right)^{2021}$$

$$= \sum_{r=1}^{\infty} \sum_{k=0}^{2021} (-1)^k \frac{{}^{2021}C_k}{2^{(k+1)r}}$$

$$\Rightarrow \sum_{k=0}^{2021} (-1)^k {}^{2021}C_k \sum_{r=1}^{\infty} \frac{1}{2^{(k+1)r}}$$

$$= \sum_{k=0}^{2021} (-1)^k \frac{{}^{2021}C_k}{2^{k+1} - 1}$$

$$= 1 - \sum_{k=1}^{2021} (-1)^{k-1} \frac{{}^{2021}C_k}{2^{k+1} - 1}$$

$$\Rightarrow q = 1 - p \Rightarrow p + q = 1$$

8. Ans (A,B,C,D)

(A) $\vec{a}$ and $\vec{b}$ are linearly dependent vectors

$$\frac{f(x)}{x^2 + x} = \frac{3}{x} \Rightarrow f(x) = 3(x + 1)$$

$$\text{Let } g(x) = f(x) - 3(x + 1)$$

$$g(-1) = f(-1) - 0 = f(-1) > 0$$

$$g(0) = f(0) - 3 < 0$$

So at least one root of $g(x)$ lies between $(-1, 0)$

$\Rightarrow \vec{a}$ and $\vec{b}$ are linearly dependent

$$(B) \frac{\int_0^{1-x} f(t) dt}{x} = \frac{3}{2}$$

$$2 \int_0^{1-x} f(t) dt = 3x$$

$$\text{Let } g(x) = 2 \int_0^{1-x} f(t) dt - 3x$$

$$g(0) = 2 \int_0^1 f(t) dt - 3(0) > 0$$

$$g(1) = 0 - 3(1) < 0$$

So $\vec{a}$ and $\vec{b}$ are linearly dependent

$$(C) \frac{\int_0^{1-x} f(t) dt}{x} = \frac{2}{3}$$

$$\text{Let } g(x) = 3 \int_0^{1-x} f(t) dt - 2x$$

$$g(0) = 3 \int_0^1 f(t) dt > 0$$

$$g(1) = -2 < 0$$

So $\vec{a}$ and $\vec{b}$ are linearly dependent

$$(D) \frac{f(x)}{x^2} = \frac{4}{3}$$

$$\text{Let } g(x) = 3f(x) - 4x^2$$

$$g(0) = 3f(0) > 0$$

$$g(1) = 3f(1) - 4 < 0$$

So $\vec{a}$ and $\vec{b}$ are linearly dependent vector.

9. Ans (A,B,C,D)

$$(A) 2 \cdot 3 \cdot 4 \begin{vmatrix} \alpha & \beta & \gamma \\ \beta & \gamma & \alpha \\ \gamma & \alpha & \beta \end{vmatrix}$$

$$= -24 (\alpha^3 + \beta^3 + \gamma^3 - 3\alpha\beta\gamma)$$

$$= -24 (\alpha + \beta + \gamma)(\alpha^2 + \beta^2 + \gamma^2 - (\alpha\beta + \beta\gamma + \alpha\gamma))$$

$$= -24 (-P)((-P)^2 - 3(0))$$

$$= 24P^3$$

$$(B) A = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} & -\frac{2}{3} \\ \frac{x}{3} & \frac{2}{3} & \frac{y}{3} \end{bmatrix}$$

$$A' = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & \frac{x}{3} \\ \frac{2}{3} & \frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & -\frac{2}{3} & \frac{y}{3} \end{bmatrix}$$

$$AA' = \begin{bmatrix} 1 & 0 & \frac{x}{9} + \frac{4}{9} + \frac{2y}{9} \\ 0 & 1 & \frac{2x}{9} + \frac{2}{9} - \frac{2y}{9} \\ \frac{x}{9} + \frac{4}{9} + \frac{2y}{9} & \frac{2x}{9} + \frac{2}{9} - \frac{2y}{9} & \frac{x^2}{9} + \frac{4}{9} + \frac{y^2}{9} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\frac{x}{9} + \frac{4}{9} + \frac{2y}{9} = 0 \quad x + 2y = -4$$

$$\frac{2x}{9} + \frac{2}{9} - \frac{2y}{9} = 0 \quad x - y = -1, \text{ then}$$

$$y = -1$$

$$x + 1 = -1$$

$$x = -2$$

$$x + y = -3$$

$$(C) AA^T = I$$

$$|A| = \pm 1$$

$$\Rightarrow |A| = 1$$

$$|A - I| = |A - AA^T| = |A| |I - A^T| = -|A - I|$$

$$\Rightarrow |A - I| = 0$$

(D) $A - A^T$ is skew symmetric odd order matrix

$$\text{So, } |A - A^T|^{2023} = 0$$

10. Ans (A,B,D)

$$\frac{x^2}{\lambda^2} - \frac{y^2}{(\lambda^3 + \lambda^2 + \lambda)^2} = 1$$

$\therefore y = mx + c$ be tangent to hyperbola then

$$c = \sqrt{\lambda^2 m^2 - (\lambda^3 + \lambda^2 + \lambda)^2}$$

$$\Rightarrow m^2 = \frac{(\lambda^3 + \lambda^2 + \lambda)^2 + c^2}{\lambda^2} \Rightarrow m^2 = (\lambda^2 + \lambda + 1)^2 + \frac{c^2}{\lambda^2}$$

$$\Rightarrow 16m^2 = 16(\lambda^2 + \lambda + 1)^2 + \frac{16c^2}{\lambda^2} \Rightarrow \min$$

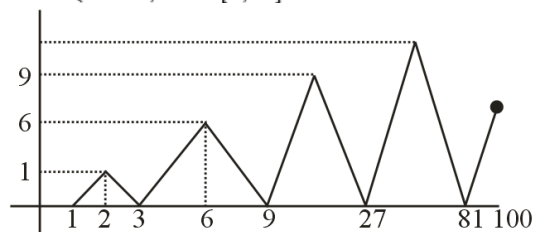
$$\text{value} = 16\left(\frac{3}{4}\right)^2 + 0 = 9$$

PART-3 : MATHEMATICS

SECTION-III

1. Ans (8)

$$f(x) = \begin{cases} x-1, & x \in [1, 2] \\ 3-x, & x \in [2, 3] \end{cases}$$



$f(x)$ is non-differentiable at

$$x = 2, 6, 12, 20, 30, 42, 56, 72$$

2. Ans (1)

As, we know that

$$\text{So, } \sin^{-1}x + \cos^{-1}y + \sec^{-1}z$$

$$\leq \frac{\pi}{2} + \pi + \pi = \frac{5\pi}{2}$$

Also,

$$t^2 - \sqrt{2\pi}t + 3\pi = \left(t - \sqrt{\frac{\pi}{2}}\right)^2 + \frac{5\pi}{2} \geq \frac{5\pi}{2}$$

Hence, the given in equation exists if equality holds, i.e.,

$$\text{L.H.S.} = \text{R.H.S.}$$

$$= \frac{5\pi}{2} \Rightarrow x = 1, y = -1, z = -1, t = \sqrt{\frac{\pi}{2}}$$

$$\text{Now, } \tan^{-1}x + \tan^{-1}y + \tan^{-1}z + \tan^{-1}\left(\frac{\sqrt{2}}{\sqrt{\pi}}t\right) = 0$$

$$\text{So, } \sec(0) = 1$$

3. Ans (1)

$$\begin{aligned} y &= \frac{\left[x + \frac{1}{x} - 1\right]^3}{x^3 - 1 + \frac{1}{x^3}} \\ &= \frac{\left(x + \frac{1}{x} - 1\right)^3}{x^3 + \frac{1}{x^3} - 1} \\ &= \frac{\left(x + \frac{1}{x} - 1\right)^3}{\left(x + \frac{1}{x}\right)^3 - 3x\frac{1}{x}\left(x + \frac{1}{x}\right) - 1} \\ &= \frac{\left(x + \frac{1}{x} - 1\right)^3}{\left(x + \frac{1}{x}\right)^3 - 3\left(x + \frac{1}{x}\right) - 1} \end{aligned}$$

$$x + \frac{1}{x} = t$$

$$y = \frac{(t-1)^3}{(t^3 - 3t - 1)}$$

$$y \text{ is min at } t = 1 + \sqrt{3}$$

$$y_{\min} = A = \frac{\sqrt{3}}{2 + \sqrt{3}}$$

$$\text{So, } \frac{2 + \sqrt{3}}{\sqrt{3}} A = 1$$

4. Ans (7)

Number of required hexagon

$$\frac{n}{6} \times {}^{n-7}C_5 = 49$$

$$\text{then, } n = 14$$

5. Ans (2)

$$\begin{aligned} \frac{\cos \frac{x}{2} - \cos\left(n + \frac{1}{2}\right)x}{2 \sin \frac{x}{2}} &\leq \frac{\cos \frac{x}{2} + 1}{2 \sin \frac{x}{2}} \\ &= \frac{\cot \frac{x}{4}}{2} \leq \frac{\sqrt{3}}{2} \end{aligned}$$

6. Ans (5)

$$A.M = \left(a + \left(\frac{b}{2} + \frac{b}{2} \right) + \left(\frac{c}{3} + \frac{c}{3} + \frac{c}{3} \right) + \left(\frac{d}{4} + \frac{d}{4} + \frac{d}{4} + \frac{d}{4} \right) + \left(\frac{e}{5} + \frac{e}{5} + \frac{e}{5} + \frac{e}{5} + \frac{e}{5} \right) \right) / 15 = 1$$

$$GM = 1$$

$$\text{So } A.M. = G.M.$$

$$\text{So } a = \frac{b}{2} = \frac{c}{3} = \frac{d}{4} = \frac{e}{5}$$

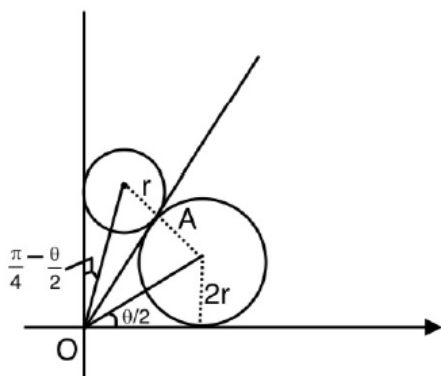
$$\text{So } a = 1, b = 2, c = 3, d = 4, e = 5$$

$$\text{So } \frac{a^2 + b^2 + c^2 + d^2 + e^2}{11} = \frac{55}{11} = 5$$

7. Ans (2)

$$OA = r \cot \left(\frac{\pi}{2} - \frac{\theta}{2} \right) \text{ and } OA = 2r \cot \frac{\theta}{2}$$

$$\text{Let, } \tan \frac{\theta}{2} = t$$



$$\Rightarrow \frac{1+t}{1-t} = \frac{2}{t} \Rightarrow t^2 + 3t - 2 = 0$$

$$\Rightarrow t = \frac{-3 \pm \sqrt{17}}{2}$$

$$\therefore \tan \frac{\theta}{2} = \frac{\sqrt{17} - 3}{2}$$

$$\Rightarrow a + b + c = 17 + 3 + 2 = 22$$

8. Ans (3)

Let a,b,c are of type $7\lambda, 7\lambda + 1, 7\lambda + 2, 7\lambda + 3, 7\lambda + 4, 7\lambda + 5, 7\lambda + 6$

$$(7\lambda)^2 \rightarrow 7\lambda \text{ type, } (7\lambda + 1)^2, (7\lambda + 6)^2 \rightarrow 7\lambda + 1$$

$$\text{type } (7\lambda + 3)^2, (7\lambda + 4)^2 \rightarrow 7\lambda + 2 \text{ type,}$$

$$(7\lambda + 2)^2, (7\lambda + 5)^2 \rightarrow 7\lambda + 4 \text{ type. either a,b,c are}$$

of 7λ type (only one possible way) and either one

of $(7\lambda + 1), (7\lambda + 6)$ and 2nd of $(7\lambda + 2), (7\lambda + 5)$

type and third of $(7\lambda + 3), (7\lambda + 4)$ type so

$$\text{number of ways} = {}^2C_1 \times {}^2C_1 \times {}^2C_1 \times 3$$

$$\text{probability} = \frac{1 + 2 \times 2 \times 2 \times 6}{7^3} = \frac{1}{7}$$